

IEDA 4000H

Homework 3

Due Date: Nov 2 (Sunday)

1. You work in an investment fund overseeing a portfolio of \$1M, evenly distributed across five assets in the investment universe. Your boss has decided that you must rebalance the portfolio every 21 trading days according to a Markowitz portfolio with $\lambda = 0.1$:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \lambda \mathbf{w}^T \Sigma \mathbf{w} - \mathbf{w}^T \boldsymbol{\mu} \\ \text{s.t.} \quad & \mathbf{1}^T \mathbf{w} = 1, \\ & \mathbf{w} \geq 0. \end{aligned} \tag{1}$$

You need to test the strategy proposed by your boss. The mechanism you need to implement is called walk-forward (WF) backtesting based on rolling windows. We divide the trading days into rolling windows of 21 denoted as $[T_1, \dots, T_{21}]$, and in each T_n we will stick with the portfolio in (1) using the sample mean and sample covariance of the previous 21 days as $\boldsymbol{\mu}$ and Σ . At the start, the portfolio is an equally weighted portfolio (EWP), as on the first window $\boldsymbol{\mu}$ and Σ can not be estimated due to missing data.

- (a) Using [yfinance](#), get the historical prices at close from the **13th of March 2020** to the **16th of May 2024** for the following tickers: AAPL, MSFT, AMZN, JPM, TSLA. Compute the log returns and use them for the backtest.
- (b) Compute the portfolio matrix $\mathbf{W} = [\mathbf{w}^1, \dots, \mathbf{w}^{21}]$ and print it. Plot the Net Asset Value (NAV) of the portfolio over the data history.

Rebalancing involves executing trades, which incur transaction costs diminish the overall return of the investment and typically come from two sources: fees & commissions (brokerage, exchange, etc.) and slippage.

- (c) Considering a transaction cost rate of 25 bps on both sides of the trade, that is, for each dollar value of asset bought/sold you pay 0.0025 dollars, calculate and print the total transaction costs incurred for the whole backtest. Overlay a new NAV plot taking into account transaction costs on the last plot.
(Basis points, denoted by bps, represents a factor of 10^{-4} . 1 bps means 0.0001 or 0.01%)

As you can observe in (c), transaction costs have an impact on the performance of a rebalancing strategy. Therefore, let's take them into account each time we rebalance the portfolio. Because the exact transaction cost is usually not known until the trade is executed, it is common to approximate it as being proportional to the portfolio turnover. That is, the transaction cost is modeled as

$$\gamma \|\mathbf{w} - \mathbf{w}^{op}\|_1, \tag{2}$$

where $\mathbf{w}^{op}$ is the original position and γ is the proportionality constant. The value of γ depends on the transaction cost rate, but also on the time horizon and rebalancing frequency. For simplicity, let's take $\gamma = 1$.

- (d) Formulate the Markowitz portfolio as in (1) but now including a penalty term related to transaction costs as in (2). Using this formulation for the rebalancing, compute the new portfolio matrix $\mathbf{W} = [\mathbf{w}^1, \dots, \mathbf{w}^{21}]$ and print it. Calculate and print the total transaction costs incurred on the whole backtest and compare with question (c). Overlay the NAV of the portfolio over the plot from (c). You must solve the optimization problem for rebalancing using [qpsolvers](#).
- (e) Compute the annualized Sharpe ratio, annualized return and annualized maximum drawdown for the two portfolios when taking into account transaction costs (that is, the ones from part (c) and (d)). Assume a risk-free rate of 2%.

2. Find the conjugate for each of the following functions:

(a) $f(x) = \max\{0, x\}, \quad x \in \mathbb{R}.$

(b) $f(\mathbf{x}) = \ln \left(\sum_{i=1}^n e^{x_i} \right), \quad \mathbf{x} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n.$

3. Proof the following properties:

- (a) (Fenchel's inequality) For any function $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$, show that for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$f(\mathbf{x}) + f^*(\mathbf{y}) \geq \mathbf{x}^\top \mathbf{y}.$$

- (b) For any function $f : \mathbb{R}^n \rightarrow [-\infty, +\infty]$, show that for any $\mathbf{x} \in \mathbb{R}^n$,

$$f^{**}(\mathbf{x}) \leq f(\mathbf{x}),$$

where f^{**} is the biconjugate of f , defined as

$$f^{**}(\mathbf{x}) = \sup_{\mathbf{y} \in \mathbb{R}^n} \{\mathbf{x}^\top \mathbf{y} - f^*(\mathbf{y})\}.$$